Worksheet 3-4 A
Differentiating Inverse Trigonometric Functions #1
AP Calculus Name_____
YOU MUST SHOW ALL WORK!

Directions for 1 – 8: Differentiate each function with respect to x.

1.) $y = \cos^{-1}(-5x^3)$

2.) $y = \arcsin(-2x^2)$

3.) $y = \arctan(2x^4)$

4.) $y = \csc^{-1}(4x^2)$

5.) $y = (\arcsin(5x^2))^3$

6.) $y = \cos^{-1}(-2x^3 - 3)^3$

7.) $y = \sec^{-1}(3\ln x)$

8.) $y = (\tan^{-1}(5x))^2$

Directions: Find the equation of the tangent line to the curve at the given point.

9.) $y = \arcsin(2x)$ at the point where $x = \frac{1}{4}$

10.) $f(x) = \arccos(3x)$ at the point where $x = -\frac{\sqrt{3}}{6}$

11.) $y = \arctan(\sqrt{x})$ at the point where $x = 3$